18

Dealing with Limitations

No plan survives contact with the enemy.

—HELMUT VON MOLTKE THE ELDER

No research is perfect. Every study has limitations, and every data set has blemishes. You have to address these even while focusing on meaningful results and that understanding that grows from them. The question is: how do you address the negatives without undermining the positives?

Many writers address limitations using what I call a "yes, but" strategy, where they present the results and conclusions and then discuss their limitations. The most classic form of this is the "more research is needed" resolution. The problem with "yes, but" is that it puts the "but" last, making it the resolution and most powerful message.

So what is the alternative to "yes, but"? Turn it around. Instead of "yes, but," use "but, yes." Deal with the limitations as early as possible, get that discussion out of the way, and then get on with developing a strong story. Constrain your conclusions to fit within the limitations but end with a "yes."

"But, yes" allows you to end powerfully, making for better storytelling. Importantly, though, it is also easier for the reader to follow and, I believe, more honest. It achieves these aims by airing your dirty laundry up front, rather than

having it feel like the fine print in an advertisement, something you hope readers will miss. Having laid out the limitations, readers can work through your data and discussion in light of them. They never have to back up and reprocess information—that takes work and creates confusion. Depending on the issues, you may need to address limitations in any part of a paper or proposal; as a rule, earlier is better.

18.1. THE INTRODUCTION: PROMISE THE STORY YOU WILL DELIVER

Many problems arise not from inherent limitations in the methods and resulting data but from a mismatch between the question and the methods. Authors ask a good question but use methods that are inadequate to answer it. Such problems can sometimes be resolved simply by revising the question. In the Introduction, frame the knowledge gap you will actually fill—set up expectations you can deliver on.

As an example, imagine a paper submitted to *Soil Biology & Biochemistry* in which the Introduction said the following.

> Example 18.1
> To determine the size of the bacterial population in soil, we plated bacteria on nutrient agar and counted colonies.

I would probably reject this paper without review. We've known for a century that at most 1 percent of soil bacteria grow on agar plates, so this method can't possibly determine how many bacteria there are in soil. To count total bacteria, you have to do microscopic counts or DNA analyses. If on the other hand, consider if the paper said the following.

> Example 18.2
> To determine the size of the bacterial population in soil that is able to respond to inputs of fresh, high-quality organic materials, we plated bacteria on nutrient agar and counted colonies.

For this question, the approach is at least defensible. In fact, the methods that would work for the first question (microscopy and DNA) would fail for this one. Those methods wouldn't tell you which bacteria would grow if you fed them; plating might. So this one I would send out for review.

Of course, if this were the challenge, we would have to revisit the opening to ensure they frame a knowledge gap about bacteria that respond to inputs, rather than about the total population. That redefines the research and constrains its scope, but doesn't necessarily make it less interesting or important. It might even make it more interesting—I'm fine with small insightful questions. As I've said before, it is better to fill a small knowledge gap than none at all, which is what readers will think if questions and answers don't match.

18.2. MATERIALS AND METHODS

The second place you can address limitations is in the Materials and Methods. When limitations relate to experimental details or analytical methods, discuss them immediately to lay any concerns to rest. If you don't, readers may decide the work is so flawed that there is no reason to read further, or they may be distracted enough by their concerns to miss your important points. At best, they remain skeptical and you would struggle to gain their full acceptance. You are much better off if you can address readers' concerns as soon as they arise.

An excellent example of how to do this is in a paper evaluating overwinter survival of snowshoe hares in northern Canada. One of the experimental treatments involved a large fenced exclosure to keep out predators.

Example 18.3
Our final two treatments increased hare survival by excluding predators. One 1-km^2 area was enclosed by chicken-wire and electrical fencing (predator-exclosure) from 1987 to 1996 to exclude terrestrial predators, primarily lynx (*Lynx canadensis* Kerr) and coyotes (*Canis latrans* Say).[1]

The limitation here is that the design is unreplicated—they had only one exclosure. Traditional statistical logic might argue that without replication it is impossible to determine whether effects were due to the exclosure itself. Some people might read this and think "unreplicated and unstatistical, therefore unreliable and unscientific." How to address this? The authors did so with this paragraph.

Example 18.4
The predator-exclosure and predator-exclosure + food sites were not replicated, owing to the high costs of maintaining 8 km of electric fencing on a nearly daily basis for a decade. Ecologists are caught increasingly in the tension between the manipulations necessary to understand large-scale ecological processes and the impossibility of replicating large manipulations. Our control sites sampled a wide range of natural variation within our study area, increasing the likelihood that differences on the treatments were due to our manipulations rather than to underlying natural variation.

This addressed nonreplication in a way that was effective and unapologetic. First, the authors made it clear why it would be impossible to replicate. Then they grounded this within the larger discussion of how we study ecology: we often can't use nice, tidy replicated experimental designs to study messy nature. Environmental scientists must (and do) accept pseudo-replicated experiments as long as the investigators are sensitive to and deal appropriately with their limitations. Inferences of causality have to go outside simple statistics. After all, we study

1. K. E. Hodges, R. Boonstra, and C. J. Krebs, "Overwinter Mass Loss of Snowshoe Hares in the Yukon: Starvation, Stress, Adaptation, or Artifact?" *Journal of Animal Ecology* 75 (2006): 1–13.

"earth system science" but have only one Earth. Finally and importantly, they discussed how they dealt with the limitations to convince readers that differences between control and exclosure were likely real treatment effects. Having done this, they didn't raise the issue again. They got on with discussing the real science and the important results.

Addressing methodological constraints as soon as you describe the method is even more vital in proposals than papers. When your audience is unsympathetic and critical, you can't afford to give them any excuse to "go negative." Particularly when people disagree about methods, it is important to explain why yours will answer your questions. If there are limitations, explain how you get around them. If there are things that you are not going to do, tell us why (too expensive, unnecessary for your specific questions, whatever). If you avoid mentioning the negatives, reviewers will find them anyway, criticize you for them, and probably recommend rejection. If you discuss them openly, a reviewer may still object, but you might convince them you are right, or at least that you've thought about it enough to get the benefit of the doubt.

An example of this strategy is a proposal I wrote some years ago to characterize soil organic matter (SOM) in the arctic tundra. Tundra soils contain a huge stock of organic C that could be released as CO_2, accelerating climate warming. We wanted to find out the amount and chemical form of the SOM that is biodegradable. The challenge was that SOM is comprised of a complex mix of molecules, ranging from fresh plant sugars to ancient humified gunk. Some of these are easy to process, whereas others are effectively inert. There are no methods for unequivocally characterizing these materials. There are, however, several methods that give insight into specific aspects of SOM chemistry.

The first time I submitted the proposal, I described the methods without discussing their challenges. Reviewers hit on the methods' limitations, and the proposal was rejected. In the resubmission, I included the following section (slightly condensed).

Example 18.5
Potential pitfalls

Each of the methods proposed has potential limitations, and the utility of several have been questioned. However, we believe that we have constrained the limitations adequately. We have done this by being careful in interpreting what the data actually tell you (e.g. isotope equilibration as a method for estimating microbial metabolites rather than the total active fraction) and in part by carefully integrating different approaches to complement each other. Thus we believe that this suite of methods will do a good job of characterizing the nature of available C and N across the tundra landscape.

The isotope equilibration technique relies on several assumptions. As originally construed, it required the assumption that it labels the active fraction without labeling the recalcitrant fraction. This is debatable since some microbial products are recalcitrant. However, the assumption that it

labels the microbial metabolite pool is substantially more robust. There is rarely extensive abiotic immobilization of either $^{15}NH_4^+$ or glucose, so the isotopes should be in the metabolite pool.

The utility of chemical fractionation in characterizing soil organic matter has been debated ever since the first fractionations of fulvic and humic acids. Chemical fractionation of *plant material*, however, has been quite successful in characterizing litter quality. Since in tundra soils much of the organic matter is still plant detritus these techniques should be useful.

The second time around, the proposal was funded. Reviewers still commented on the methods and the challenges of the project, but enthusiastically, recognizing that we had balanced methods' strengths and limitations and we had thought about how to interpret the results. We didn't change the methods, just how we defined their results.

I sometimes say that there is no such thing as a bad method. As long as it is done well, a method gives solid data and real information. The "bad" part always relates to interpretation—a method may truly be bad for measuring one thing but good for something slightly different. In Example 18.3, the isotope equilibration technique was bad for estimating the active fraction, but good for estimating microbial metabolites. Make sure to explain why your methods give the information you need.

18.3. DISCUSSION

When constraints are methodological detail, address them in the Methods. When they go beyond that to affect how you interpret the data, you must address them in the Discussion. You must openly discuss limitations but without highlighting them so strongly that they become an argument for rejecting the paper entirely. Doing this involves both story structure and language.

In terms of structure, the question is where to put the discussion of limitations. Most of the time, you should avoid the power positions of the Discussion's opening and resolution. Those are for critical story points, and if limitations are the critical point, why submit the paper? You should generally find some convenient place early in the body of the Discussion to discuss the work's limitations and constraints.

As for language, do it briefly and honestly, then get on with the story. Example 8.7 used this approach. The authors included two constraints on the conclusions, but they put them in the middle of an LD paragraph—after the lead but before the rest of the discussion.

A good discussion of methodological limitations comes from a paper that explored the side effects of inhaled cortical steroids (ICS) as a treatment for asthma. Clinical trials may underestimate side effects because trials are usually short and have restricted selection criteria for who participates. To better evaluate side effects of ICS, the researchers surveyed people who routinely used them and found effects ranging from gummy mouths to mood swings and sleeping problems, some of which had not been well documented. But survey-based studies always have limitations that the authors rightfully felt they needed to bring to readers' attention.

Example 18.6

Limitations of our research include potential selection bias. Although our response rate (56%) is not atypical for a questionnaire study, we acknowledge that responders may have had a greater propensity for perceiving ICS side effects as compared to non-responders, and it is possible that side effect reporting may have been inflated in all dose groups as a result. Alternatively, ICS users in our study may have experienced relatively mild ICS side effects, since they continue to use ICS regardless of their perception of side effects.[2]

This pointed out that there is some quantitative uncertainty about the results that can never be overcome in a survey study—you can't make people answer, or answer honestly. But there were robust patterns in the results, offering information that physicians should be aware of, which the authors then elaborated.

Another example of dealing with limitations in the Discussion comes from a paper about how variation between individuals can affect transcription patterns in different tissues, ultimately leading to different physiology.

Example 18.7

Despite the limited number of validated examples, osteoblast-specific transcript isoforms are present and indicates that expression of certain alternatively spliced variants in bone tissue are controlled by different regulatory SNPs and vary between individuals and/or populations, with potentially important biological functions.[3]

The authors start with the opening clause "Despite the limited number of validated examples," which frames the "but"; they then give the "yes" by saying that osteoblast-specific transcript isoforms *are* present and suggest the implications of that. They use the "but, yes" approach within a single sentence. They remind us of the limitations but don't dwell on them.

2. J. M. Foster, L. Aucott, R. H. W. van der Werf, M. J. van der Meijden, G. Schraa, D. S. Postma, and T. van der Molen, "Higher Patient Perceived Side Effects Related to Higher Daily Doses of Inhaled Corticosteroids in the Community: A Cross-Sectional Analysis," *Respiratory Medicine* 100 (2006): 1318–36.

3. Kwan et al., "Tissue Effect on Genetic Control of Transcript Isoform Variation," *PLoS Genetics* 5 (2009): e1000608. DOI:10.1371/journal.pgen.1000608.

Although you should generally avoid raising constraints and limitations in power positions, on rare occasions it is necessary. When experimental approaches are unusual or novel, it can be important to give readers some calibration on their strengths and weaknesses before discussing the results themselves. In these cases, it is useful to open the Discussion with a section that weaves the methods considerations into the opening of its story arc.

An example of this approach is a paper I wrote evaluating controls over nitrogen movement between decomposing leaves in forest litter. As leaves fall, they form a layer on the soil surface, a layer that might include leaves from different tree species, some of which are N-rich while others are N-poor. This creates a complex environment in which the movement of N from rich to poor leaves may regulate the decomposition of the whole litter layer.

The experiment used an ^{15}N isotope tracer to follow N movement from ^{15}N-labeled leaves to unlabeled leaves, but it used the "microcosm" design shown in figure 18.1: two leaves (one labeled, one not) held together with wire and wrapped in plastic to prevent them from drying out—not quite what happens in an actual forest. Because it was a horribly artificial experimental design, we felt it important to be explicit about the constraints this placed on data interpretation, and we put that discussion in a prominent place that a reader couldn't miss. We started the discussion with an "Experimental Considerations" section.

Example 18.8

Experimental Considerations

This experiment used a rather artificial structure: pairs of individual leaves held together to ensure contact and minimize diffusion limitations. We labeled [with ^{15}N] only the microorganisms growing on the litter and so fungi were likely involved in decomposing the leaves, we did not observe fungal mycelia developing between them; such mycelia are thought to be important in moving N through a natural litter layer. Thus, this design provides little information about the amount of N that would actually move within natural litter layers decomposing *in situ*. However, this controlled experimental design allowed us to do a ^{15}N budget on each individual leaf-pair microcosm and thereby evaluate the bi-directional N movement among leaves of different species and N-levels, and to evaluate the separate role of source and sink N concentrations in regulating N movement within the microcosms.[4]

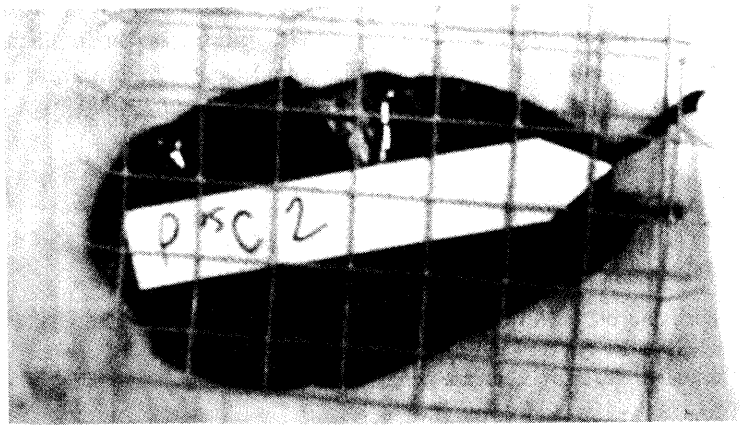

Figure 18.1. The set-up for an experiment evaluating N movement between decomposing leaves.

This was honest about the limitations of the experiment and that it told us little about the rates of N-movement in the field. But it provided a venue for arguing that we could assess mechanisms that controlled movement and that this offered a useful insight. This discussion also allowed us to remind readers of the knowledge gap we were targeting and the questions we were asking.

18.3.2. Within the Conclusions

You should never make limitations *the* conclusion, but sometimes you may need to mention them *within* the conclusions. In such cases, you need a tightly defined "but, yes" structure to frame the limitations as quickly as possible and then to push on to the conclusions—what you learned.

A particularly clever approach to doing this is illustrated in example 18.9. This is from a paper that evaluated approaches to estimating the mortality of children under five years old in nations that don't keep records of births and deaths. There was no absolute data set to use as a reference to assess the reliability of the estimates. That left uncertainties that constrained the conclusions. But if the authors had included that entire discussion in the concluding paragraph, it would have overwhelmed it. Instead, they discussed the limitations in the next-to-last paragraph, which was long and involved (21 lines). The final conclusions paragraph was only 11 lines.

Example 18.9

Despite these limitations, the methods proposed here represent a major advance on current practice and offer the prospect of vastly increasing our knowledge about levels, recent trends, and inequalities in child mortality. If we are to make rapid progress with the unfinished agenda of reducing child deaths, policy and practice must be better informed by more comprehensive,

4. J. P. Schimel and S. Hättenschwiler, "Nitrogen Transfer between Decomposing Leaves of Different N Status," *Soil Biology & Biochemistry* 39 (2007):1428–36.

> relevant, and timely information. Systematic application of the methods proposed here will establish that evidence base, and thereby increase accountability among countries and the global health community to accelerate efforts to reduce the global toll of child deaths.[5]

This reminds us of the constraints, but with a tidy nominalization ("these limitations") that refers back to the previous paragraph. This simultaneously highlights and minimizes them! By separating the negatives and the positives into separate story arcs and using effective language to link them together, the authors addressed the constraints honestly in a way that did nothing to undermine the power of their conclusions. This was nicely done.

As I said at the beginning of this chapter, no research is perfect, and there is nothing to be embarrassed about in admitting it. Rather, the opposite is true—it's our responsibility to address our work's limitations. The "but, yes" approach does this in a way that is open about the limits but highlights the conclusions. It makes the presentation clear, strong, and credible. By clarifying both the limits and the power of your work, it will motivate others to pick up and build from it, that is, to cite it.

5. J. K. Rajaratnam, L. N. Tran, A. D. Lopez, and C. J. L. Murray, "Measuring Under-Five Mortality: Validation of New Low-Cost Methods," *PLoS Medicine* 7 (2010): e1000253. DOI:10.1371/journal.pmed.1000253.

Writing Global Scien

Life isn't fair.

Science is increasingly dominated by scholars for whom English is a secon
guage and by nations with developing scientific cultures. These don't nece
overlap: Germany was a founder of the modern scientific tradition, whereas
where English is well established, has a developing science program. Lar
and scientific culture, however, can each pose challenges to publishing
international marketplace of high-impact journals.

Many people understandably feel language is *the* struggle. Writing a sci
paper seems daunting when even ordering dinner at a restaurant can be
If English is your second language, you may feel that writing science must b
for native speakers. If that were true, I wouldn't have written this book. Wri
hard for all of us.

The hardest part of writing science, though, is developing the story and
it out cleanly. The essence of getting the story across is structure: knowing w
put where. Structure comes before language in the SCFL formula I discus
chapter 17, and most of this book is about structure. Story structure tran
language; OCAR isn't about English.